Supplementary Information

TaoChronos: A Reliable Agent for Knowledge Discovery in Traditional Chinese Medicine

TaoChronos (道乃久)

Contents: Supplementary Notes 1–4; Supplementary Tables 1–12 (each also as CSV in `tables/`); Supplementary Fig. 1; Supplementary Data 1 (the Discovery Report of the end-to-end run) and 2 (its event log).

Supplementary Note 1 | Evaluation design

TaoChronos-Eval has forty suites in three layers. The first tests the harness's boundaries: nineteen invariants run as probes against the harness under test, each checked by mutation (the boundary removed, the invariant broken, the boundary restored; Fig. 4b measures all nineteen on every variant). The second tests readings: 107 verbatim quotations that can still be read wrongly, in eight classes, with unsafe errors (those that would let a false claim through) counted apart. The split into 23 development and 84 held-out cases was fixed before the suite first ran, and four blind rounds were each written before the rules that first read them (Fig. 4d). The third tests research value: a blind temporal benchmark in which the store is split at a cutoff year, passages that may have been written on either side belong to neither side, the rules were written before the first run (evals/gold/research_value.yaml) and every read path is bounded in SQL, so that leakage is counted rather than assumed; graph rankings are compared with popularity, chance and two other link predictors, and the pipeline's own predictions are judged by later books.

The other suites measure each capability in the way its task allows: constructed data where an answer can be constructed (planted drug pairs, artificial manuscript traditions, a lost work quoted by four books), a development set where rules need examples, and results the literature already holds where the real corpus can be checked (the taboo renamings of Fig. 6, the Yunqi chapters of the Suwen, the first gloss of 几几). Development and constructed sets were written with the rules they test; they are regression tests of intended behaviour, and only held-out or real-corpus results estimate generalization (Supplementary Fig. 2). The two held-out sets of the V2 layer (entities, ancient phenotypes) were annotated before the rules ran on them. Everything but the full-corpus suites reruns on the demonstration corpus in under a minute (`taochronos eval`); the full-corpus values, which need a store that is not distributed, are transcribed from docs/evals.md and marked. Supplementary Table 1 lists every suite.

Supplementary Table 1 | The forty evaluation suites: question, data and split, metrics, result and where the paper reports them (`tables/S1_evaluation_design.csv`)

Layer	Suite	Question	Data and split	Metrics	Result	Reported in
Boundaries	invariants	Do the harness's boundaries hold — scope on every read path, semantic support at commit, objections a model cannot close, AfterToolCall denials, isolation of model code, atomic batches, G7 for experts only, evidence bounds, G8 by stance and level, read-only agent views, integrity, retraction, publication on current versions, acceptance ≠ execution?	19 probes (the audit's counterexamples P01–P10 and nine boundaries added with them) run against the harness under test; six boundaries removed in turn (mutation)	held / exercised / broken	19/19 held (P01–P10 and nine boundaries added with them); each of six boundaries removed on purpose (scope guard, result firewall, semantic support hook, reducer authority, evidence bounds, publish guard) breaks its invariant	Fig. 4b; Table 2
Readings	interpretive	Are true quotes read right where the reading is hard — inverted relations, a prohibition beside a prescription, consequences, reported and rejected views, omitted conditions, mixed witnesses, disputed dates, textual cruxes?	107 verbatim cases in 8 classes (gold/interpretive.yaml); 23 development, 84 held-out, split fixed before the first run; 4 blind rounds	accuracy and unsafe errors, per class and per split (dev / held-out)	held-out 82/84 (0.976), 2 unsafe (大黄牡丹汤 read as an indication; 里水, a crux the harvest does not find); dev 23/23. First blind round 43/53 (0.811); round 3, before the crux registry, 7/10 with 3 unsafe cruxes; see below	Fig. 4c,d; Table 2
Research value	research_value	On the full corpus split at a cutoff year: do	full corpus split at 1127 and 1368: 230 topic–cutoff pairs read, 189 with	hits and precision@k, mean average	precision@10 0.168 (popularity 0.134; chance 0.054);	Fig. 5; Supplementary Fig. 3; Table 2

Layer	Suite	Question	Data and split	Metrics	Result	Reported in
		the graph rankings D4 uses anticipate the remedies later books link to a disease, better than popularity and chance? Do the pipeline's own predictions, made on the past alone, come true?	a link that came true, 29,467 candidates; the pipeline run on the past view at 1368 for 2 development and 4 held-out topics; replicated on the open corpus (Supplementary Fig. 3)	precision, paired sign tests over topics; predictions held (strict / loose), per split and review status; leakage	average precision above popularity in 149 of 189 pairs ($P = 2.4 \times 10^{-16}$); held-out predictions: discoveries 7/19; open corpus: above popularity 24/6 ($P = 0.0014$), not separable from chance 13/17 ($P = 0.58$)	
Verification	philology	Are contested readings kept with probabilities and an uncertain mass? Are variant spellings normalised for matching?	7 contested readings (gold/philology.yaml)	pass rate	7/7	—
Verification	claims	Does extraction find the right hyperedges (relation + key arguments + negations)?	42 gold hyperedges (gold/claims.yaml), development	precision · recall · F1 · recall by relation · provenance pass rate	P 1.0 · R 0.976 · F1 0.988 (miss: the second 湿痹 sign definition). Before the general PROGNOSIS rule (则愈 / 自愈 / 欲解 / 可治 … / 者死 / 难治 / 不治 …): P 0.976 · R 0.952 · F1 0.964, prognosis recall 0 — this gold is dev	Table 2
Verification	provenance	Are all committed claims and evidence verbatim? How deep is the Claim → Pixel chain?	every claim and evidence record a demonstration run commits	verbatim rates · chain depth · level availability	100% verbatim claims and evidence; mean chain depth 7.8; page/line/region/image unavailable (no images in the demo)	Table 2
Verification	hallucination	Are fabricated, one-character-perturbed and mis-positioned quotes caught, without flagging real quotes or variant spellings?	60 random passages × 4 probes: verbatim, one character replaced, random characters, shifted span	detection rates · verbatim recall	fabricated 100% · perturbed 100% · wrong span 100% detected; verbatim recall 100%; variant spelling accepted	Fig. 4a; Table 2
Verification	contradiction	Contradiction vs conditional vs apparent vs support vs unrelated (including differential diagnosis and transcription)	12 passage pairs (gold/contradictions.yaml)	accuracy · macro-F1 · confusion	12/12	—
Verification	lineage	Formula derivation, citation, transcription, rephrasing, inheritance, opposition	typed links of the demonstration corpus (gold/lineage.yaml)	precision (judged) · recall · unjudged predictions	P 0.867 · R 0.929 — false positives 逍遥散 ← 苓桂术甘汤 and 补阳还五汤 ← 透脓散 (convergent compositions); miss: 王清任's implicit opposition	—
Verification	temporal	Period constraints, earliest attestations (through variant spellings and concepts that predate their names), period leakage	6 period constraints, 6 earliest attestations (gold/temporal.yaml)	accuracies · leakage count	constraints 6/6 · earliest 6/6 · leakage 0	Table 2
Verification	anachronism	Are equations of historical and modern concepts blocked, careful phrasings allowed, mappings typed conservatively, agent-asserted equivalence refused?	10 statements, 8 mappings, agent-asserted equivalences (gold/anachronism.yaml)	statement accuracy · block P/R · mapping accuracy	statements 10/10 · mappings 8/8 · agent equivalence blocked	Table 2
Verification	recovery	After crashes at several points, does resume reach the identical scientific state?	crashes at 4 points of a run; reruns; parallel against sequential runs	success rate · replay consistency	4/4 crash points; deterministic rerun; parallel = sequential; replay consistent	—

Layer	Suite	Question	Data and split	Metrics	Result	Reported in
		Are reruns and parallel runs identical?				
Research value	time_machine	Research on texts before a cutoff (a separate harness, nothing later indexed); are its structured predictions borne out by later texts?	the demonstration corpus split at 1368 (a separate harness indexes the past only)	decidable predictions · accuracy (survived vs other) · leakage · link-prediction baselines	a passage is on one side of the cutoff only when its whole date range is (straddling texts are in neither the past corpus nor the checking future; none at 1368); 3 decidable predictions, 2 held (taboo rename persisted; lost association stayed absent); failed: the 肾气丸-family core predicted from two members — 张景岳's 左/右归丸 dropped 泽泻·牡丹皮·茯苓	—
Research value	source_rediscovery	Hold out a cited book: is it re-inferred as a missing source, dated before its earliest citer?	3 cited books held out (gold/rediscovery.yaml)	recall@3 · date-bound consistency	3/3 re-inferred with consistent date bounds	—
Verification	ablations	Which components carry the results?	demonstration corpus: 6 terms, 12 passage pairs, the agents' context, runs without the Skeptic	earliest accuracy without routes · contradiction accuracy without sense/philology · G5 inflation without clusters · Context-OS token reduction · false survivals without the Skeptic	earliest accuracy: all 1.0 · no sense route 0.67 · BM25 only 0.67; contradiction: no sense resolution 0.83 · no philology 0.92; 1 hypothesis would pass G5 only by counting dependent books; Context OS −94% tokens; without the Skeptic 1 rejected hypothesis survives and no revisions happen	Supplementary Fig. 2c
Deep discovery	collation	Are the variants of a known tradition found, and its stemma, groups and contamination recovered?	5 artificial traditions copied from the demo corpus down a known stemma (two branches; one witness copied from both)	unit recall / precision · split recall · Robinson–Foulds · group precision · contamination P/R	variant units: recall 0.998 · precision 0.999; the true split recovered in every trial (RF 0.33: neighbour-joining resolves the true polytomy — three copies made independently from one exemplar — into an extra split); top-3 groups all genuine (all groups 0.73); contamination P 1.0 · R 1.0	Table 2
Deep discovery	reuse	Is each reuse typed right (直接引用 … 套语相似), and are real reuses found?	33 author-constructed pairs (gold/reuse.yaml), all eight types and unrelated pairs	accuracy · macro-F1 · mode accuracy · detection P/R/F1 · pipeline recall	accuracy 0.97 · macro-F1 0.965 · mode 0.97; detection P 1.0 · R 0.96 · F1 0.98; the demo corpus's gold transcriptions and rephrasings found by study.reuse: 1.0	Table 2
Deep discovery	stratigraphy	Are the layers of composite texts found, with their boundaries, and are texts attributed to their authors?	5 composites of 16 000 characters with blocks rewritten in a second author's function characters; the real 素问 and 伤寒论	layer accuracy · change-point and boundary P/R · attribution accuracy · real checks (素问 运气七篇, 叔和 chapters)	layer accuracy 0.955; boundaries R 0.90 · P 0.93; change points R 0.45 · P 1.0; attribution 0.89 · real: the 运气七篇 7/7 in the minor layer, both of their boundaries found, 3 of the 4 chapters flagged by late vocabulary are 运气 chapters, 辨脉法 and 平脉法 nearest the 脉经	Table 2
Deep discovery	cases	Are case records cut into cases and visits, and are their fields read right?	6 constructed cases in the two manners of the literature — visits under a heading, narratives (gold/cases.yaml); the two transcriptions of 吴鞠通医案	segmentation · per-field accuracy · response and outcome accuracy · agreement of two transcriptions	segmentation, fields, responses, outcomes 1.0 on the development set · real: case counts agree 0.98, outcomes 0.92	—
Deep discovery	argument	Are the steps of reasoning found and typed?	18 author-constructed passages (gold/argument.yaml); the 脉经's two transcriptions against 温病条辨	labelled and unlabelled edge P/R/F1 · clause segmentation · same-work vs other-work divergence	edges F1 1.0 on the development set · real: JSD 0.020 between the transcriptions of one work, 0.091 against another	—
Deep discovery	senses	Is a planted shift of senses dated, and a	synthetic occurrences: one sense rises from 15 % to 75 % in the year	change-year error · detection · labelling	change year within 2.3 years; detected 1.0; labelling 0.93; the	—

Layer	Suite	Question	Data and split	Metrics	Result	Reported in
		sense no cue names found?	1000, an uncued sense appears from 1300	accuracy · hidden sense found · curated exemplars	uncued sense found 1.0 (precision 1.0); curated exemplars of 消渴 6/8	
Deep discovery	fragments	Is a lost text rebuilt from its quotations without taking in other sources' words?	a constructed lost work (集古方, held out) quoted by four books in the manners of the 外台秘要, the 医心方, the 证类本草 and a late paraphrase, beside other sources (gold/fragments/); the real 千金 and 肘后	precision · coverage · decoy leaks · volume accuracy · real verification (千金, 肘后)	precision 0.86 · coverage 1.0 · no other source's words taken in · volumes 1.0 · real: 千金 precision 0.32 · coverage 0.12; 0.876 of the 肘后's quotations not in the extant text	Supplementary Fig. 2a; Table 2
Task specification	punctuation	Are the marks of 白文 placed where editors place them, with every character kept?	the store's punctuated texts; 55 works held out of training (104 k characters); 4 Siku 白文 works against their 笈成 punctuation	boundary / sentence F1 · mark-type accuracy · preservation rate · rule baseline · held-out works · Siku 白文	held-out boundary F1 0.767 (P 0.725 · R 0.814; rules 0.235), typed sentence F1 0.543, mark types 0.805; 白文 boundary F1 0.780 (rules 0.303); characters preserved 1.0	Supplementary Fig. 2b; Table 2
Task specification	commentaries	Are the commentaries on a clause found in every layout, attributed and compared?	a constructed classic with commentaries in four layouts (gold/commentaries/); the real 太阳之为病	unit P/R/F1 · text F1 · relation F1 · quotation recall · real: known commentators and relations	units F1 1.0 · text F1 0.81 · relations F1 1.0 · quotations 1.0 · real: 25 commentaries, known commentators 11/11, 张卿子 照录 成无己 and 喻昌 承袭 方有执 found	—
Task specification	disputes	Are named views read as rejected or endorsed, and quoted words kept apart?	22 author-constructed sentences (gold/disputes.yaml)	accuracy · reject / endorse P/R/F1 · real: known controversies	accuracy 0.86 · reject P 1.0 R 0.78 · endorse P 0.8 R 1.0 · real: 3/3 known controversies (景岳 on 丹溪's 相火, 越人's 左肾右命门 …)	—
Task specification	variant_impact	Is each variant reading classed by what it changes (dose, negation, drug, finding …)?	30 readings (gold/variant_impact.yaml)	accuracy · macro-F1 · high-impact P/R/F1 · real: the review queue of the 伤寒论	accuracy 0.97 · macro-F1 0.97 · high impact P 1.0 R 0.94 · real: the 伤寒论's 288 units, the top 10 of its review queue all high-impact	—
Task specification	clause	Are the parts of a clause (condition … prognosis) found?	16 clauses of the 伤寒论 and 金匮 (gold/clause.yaml)	piece accuracy · macro-F1 · per-role F1 · segmentation · real: forms of the 伤寒论	pieces 0.977 · macro-F1 0.92 · real: 40 % of the pieces of the fullest 伤寒论 witness read by no rule (平脉法 dialogues, numbering, notes)	—
Task specification	glosses	Are the glosses of a word found, and only of that word?	16 sentences (gold/glosses.yaml)	accuracy · P/R/F1 · real: the first gloss of 几几	accuracy 1.0 · P/R 1.0 · real: 成无己's 伸颈之貌 the first gloss of 几 几	—
Task specification	pairs	Do planted drug pairs come first, and do few chance pairs pass Benjamini–Hochberg?	600 synthetic compositions with three planted pairs; the store's 66 196 compositions	planted top-3 · false significant · real: textbook pairs · 方证 links of 桂枝汤 and 麻黄汤	planted top-3 1.0 · false significant 0.0 · real: 7/7 textbook pairs significant with lift > 1.5 (桂枝—芍药 2527 compositions, lift 2.05; 麻黄—杏仁 5.08 …)	Fig. 6d; Table 2
Task specification	schools	Are a school's planted words found, and do the textbooks' doctrines come out as its own?	5 synthetic 温病 physicians and 16 other authors; the store's schools with the textbooks' doctrines (gold/schools.yaml, 44)	planted top-2 and significant · common words listed · citation shares · real: doctrine recall and attribution	planted words top-2 and significant, no common word listed · real: doctrine recall 0.73 (z ≥ 1.96), attribution 0.64	—
Task specification	names	Is the change of a drug's preferred name dated, and its names equated and completed?	20 synthetic works (薯蓣 → 山药); the dated taboo renames (gold/names.yaml)	switch interval · nested names apart · equations · candidates · real: dated taboo renames	switch interval, nested names, equation, candidate 山芋 all right · real: 2/2 (薯蓣→山药 in 1000–1200, 玄参→元参 in 1600–1800)	Fig. 6a,b
Task specification	chain	Does each witness of a passage hang from the witness it was copied from?	6 artificial traditions of 12 witnesses; with each copy regularising spellings on its own	witnesses found · parent accuracy against the most-similar baseline, with and without convergent spellings	parent accuracy 1.0 (most-similar baseline 0.97); with convergent spellings 0.94 (baseline 0.87); witnesses found 0.96–1.0	—
V2	entities	Are the entities of a sentence found and typed (24 types)?	41 dev sentences, 160 entities (gold/ner.yaml); 25 held-out sentences, 117 entities, annotated after the rules were written (gold/ner_heldout.yaml)	precision · recall · F1	dev F1 0.991 · held-out F1 0.776 (P 0.783 · R 0.769)	Supplementary Fig. 2a; Table 2
V2	places	Are physicians placed from their signatures,	15 signatures and 5 statements of native place in the forms of real prefaces and biographies, with traps	precision · recall	P/R 1.0 · 84.6 % of physicians placed · real: 291 of 761 works placed, curated places confirmed 4/4, 湿温 in	—

Layer	Suite	Question	Data and split	Metrics	Result	Reported in
		prefaces and biographies?	(gold/places.yaml); the full corpus		清 江南 49 % vs 26 % (q 0.108, not significant)	
V2	translation	Does a translation keep the original layer and add no present-day diagnosis?	24 presentism cases (equations, retrospective diagnoses, modern names a translation adds, comparisons that must pass); the demo corpus	guard accuracy · layer preserved	guard 24/24 · original layer unchanged 24/24 · doublets glossed by their commentators 3/3	—
V2	layout	Are page images read in order, with notes, margins and OCR errors located?	60 synthetic pages (vertical, 双行小注, 版心, 书眉; noisy and character-level variants); four OCR formats; 12 OCR texts with one error each	reading-order accuracy · CER · notes F1	reading order exact 0.983 · CER 0.0005 · notes F1 0.999 · formats identical · errors located 12/12	—
V2	bridge	Are the findings of a passage projected into HPO with typed relations?	20 dev sentences; 12 held-out passages drawn at random from 65 clinical works of the full corpus, annotated before the reader ran on them; 18 canonical descriptions and 2 negatives (gold/bridge.yaml)	precision · recall · F1	dev F1 0.985 · held-out P 0.938 · R 0.732 · F1 0.822, bounds 1.0 · mapping top-1 1.0, caps 3/3, guard 20/20 · 213 HPO ids valid in hp/releases/2026-09-01	Supplementary Fig. 2a; Table 2
V2	audit	Does each statement of an answer get the verdict its evidence supports?	19 claims of every kind and 3 safety texts (gold/audit.yaml)	verdict accuracy · false SUPPORTED	verdicts 1.0 · kinds 1.0 · false SUPPORTED 0 · safety 3/3	Table 2
V2	modern	Are modern records typed by design and population, and cross-level claims refused?	10 records to type, 7 claim/evidence pairs for the cross-level guard, the synthetic OMOP database (gold/modern.yaml)	typing accuracy · guard accuracy	typing 10/10 · guard 7/7 · RWD: SQL checks pass, planted association recovered, bad SQL refused	—
V2	workflow	Does the orchestrator plan the skills a question needs and no others?	the registry; 8 questions with the skills they need and must not call (gold/workflow.yaml); one end-to-end run	valid plans · rerun from the workspace	registry valid · plans 8/8 · a rerun reads every step from its workspace	—

Supplementary Note 2 | Gates, permissions and authority

Five mechanisms bound what any part of the harness can do, and they are layered so that the failure of one is caught by another (Fig. 4b). (1) Permissions: every tool needs one permission, and an agent holds only those its specification lists (Supplementary Tables 4 and 5); an agent that asks for a tool outside them is refused and the refusal is logged. (2) Hooks: deterministic guards at eight points of the lifecycle (Supplementary Table 6) refuse a tool call or result outside the research's scope, a claim whose quote is not verbatim or contradicts it, an anachronistic equation and a report on unjudged versions. (3) Event authority: every change to the research object is an event, and the reducer refuses an event its principal may not emit; 61 event types, nine patterns of permission (Fig. 1b, Supplementary Table 7). An agent proposes; only validators grade, and only a human sets G7. (4) Gates G0–G8 grade each claim, evidence record and hypothesis on its current version (Supplementary Table 2). (5) The research contract, hashed before the first task, fixes the criteria by which a hypothesis is accepted, needs an expert, stays inconclusive or is rejected, and names the judge of each (Supplementary Table 3): finishing the tasks never accepts a research. In conversation, four core agents run the same specialists as fifteen skills (Supplementary Table 8).

Supplementary Table 2 | Gates G0–G8: what each passes, warns and fails, and who sets it (`tables/S2_gates.csv`)

Gate	Name	Subject	Pass	Warn	Fail	Set by
G0	Source	claim, evidence	the book is in the store with an origin and a licence (score 0.8 when the transcription is unverified)	—	unknown book, or no origin or licence	validator
G1	Transcription	claim, evidence	a verified manual transcription, or OCR confidence ≥ 0.9	an unverified transcription, or OCR confidence 0.7–0.9	OCR confidence < 0.7	validator
G2	Philology	claim, evidence	text collated against an edition, and the span not on a contested reading	uncollated text, or a contested reading whose base reading has p ≥ 0.4	the span rests on a contested reading with p < 0.4	validator
G3	Sense	claim; hypothesis	every term resolved to one sense for its period	a homonym whose most likely sense has p < 0.6	—	validator
G4	Claim provenance	claim, evidence; hypothesis	the quote is verbatim at its span and states the relation	verbatim, but the quote does not clearly state the relation, or states	the quote is not verbatim or not where it is said to be, or it	validator

Gate	Name	Subject	Pass	Warn	Fail	Set by
				a view the author rejects	contradicts the claim	
G5	Independent sources	hypothesis	supporting evidence from ≥ 2 independent source families (the contract may ask for more)	one family	no supporting evidence	validator
G6	Falsification	hypothesis	the latest Skeptic review leaves no open critical or major objection	an open major objection	an open critical objection (pending before any review)	validator
G7	Expert	hypothesis	a named expert approves	—	a named expert rejects (pending until then)	human only
G8	Modern evidence	hypothesis (biomedical or bridge claims)	records in people support a biomedical claim; any modern record contextualizes a typed bridge	—	no modern record, or a record contradicts the claim; not applicable to historical hypotheses	validator

Supplementary Table 3 | The criteria of the research contract, each with its judge and the state it leaves when unmet (`tables/S3_contract_criteria.csv`)

Criterion	Requirement	Judge	State when unmet	H1 in Fig. 3 (observed)
min_supporting_evidence	≥ 2	validator:contract	inconclusive	pass: 2 verified supporting record(s)
semantic_support	every supporting claim borne by its quote	validator:contract	inconclusive	pass: every supporting claim is stated by its quote
source_dependence	≥ 2 independent source families, on the current version	validator:contract	inconclusive	pass: G5 pass: 2 independent source cluster(s) (need 2): baoming_ji, jc_o128
falsification_review	a skeptic review	validator:contract	inconclusive	pass: reviewed
unresolved_hard_objections	≤ 0	validator:contract	inconclusive	pass: 0 open hard objection(s)
unresolved_critical	≤ 0	validator:contract	inconclusive	pass: 0 open critical objection(s)
alternatives_addressed	every rival explanation of the discovery type excluded or stated as open	validator:contract	inconclusive	pass: all addressed; open: same_name_other_formula
current_version_gates	G0, G1, G2, G3, G4, G5, G6 on the current version (strict: G4, G5, G6)	validator:contract	inconclusive	pass: all required gates pass on this version
expert_review	an expert's approval	human	needs_expert	pending: G7 pending
falsification	not rejected	validator:contract	rejected	R1 in Fig. 3: status rejected

Supplementary Table 4 | The sixteen agents of a research: role, mode, minimum model tier, permissions, tools, output and budget (`tables/S4_agents.csv`)

Agent	Name	Role	Mode	Model tier (minimum)	Permissions	Tools	Output	Budget (tool calls · model calls)
curator	TaoChronos-Curator	Corpus Curator. Freezes scope, hold-out and coverage into the corpus manifest.	procedure	small	classics:read	classics.list_books, vision.layout, vision.compare	CorpusManifest	10 · 0
director	TaoChronos	Research Director. Plans each round and decides which specialists to spawn; never judges evidence alone.	hybrid	frontier	classics:read, kg:read, analysis:run, codemode:exec	classics.list_books, classics.search, kg.claims, analysis.temporal_profile	ResearchPlan	30 · 6
evidence	TaoChronos-Evidence	Evidence Retriever. The Evidence Ledger — verbatim, locatable, independently sourced.	hybrid	medium	classics:read, kg:read	classics.search, classics.get_passage, kg.claims, validation.verify_quote, study.cases, study.trajectories, study.argument, graph.evidence	EvidenceSet	80 · 10
evolution	TaoChronos-Evolution	Evolution Analyst. Concept timelines and formula family trees.	procedure	small	analysis:run	analysis.concept_drift, analysis.formula_evolution, analysis.temporal_profile	AnalysisSet	20 · 0
extractor	TaoChronos-Extractor	Claim Extractor. Passages to claim hyperedges with verbatim spans.	hybrid	medium	classics:read, kg:read	kg.extract_claims, kg.term, classics.get_passage, validation.verify_quote, study.clause, study.entities	ClaimSet	80 · 10
hypothesis	TaoChronos-Hypothesis	Hypothesis Generator.	hybrid	frontier	classics:read, kg:read,	kg.claims, kg.neighbors, kg.term, classics.search,	HypothesisSet	60 · 10

Agent	Name	Role	Mode	Model tier (minimum)	Permissions	Tools	Output	Budget (tool calls · model calls)
		Observations to testable, evidence-bound hypotheses; revisions after review.			codemode:exec	classics.get_passage, validation.verify_quote		
lineage	TaoChronos-Lineage	Lineage Analyst. Citation, transcription, rephrasing, formula derivation and opposition edges.	procedure	small	classics:read, analysis:run	classics.lineage, philology.collate, analysis.missing_sources, study.schools, study.chain, study.places	LineageReport	20 · 0
meta_reviewer	TaoChronos-MetaReviewer	Meta-Reviewer. Systemic weaknesses, duplicates and next-round recommendations.	hybrid	frontier	kg:read, classics:read	kg.claims, audit.claim	MetaReview	10 · 4
modern_evidence	TaoChronos-ModernEvidence	Modern Evidence Analyst. The only door between the historical and biomedical spaces (Gate G8).	hybrid	frontier	literature:read, kg:read, classics:read	literature.modern_evidence, kg.term, bridge.map, bridge.reverse, modern.search, modern.safety, rwd.ask	ModernEvidenceSet	30 · 6
ontologist	TaoChronos-Ontologist	Ontologist / Terminology Curator. Typed historical-to-modern mappings, proposed only.	hybrid	frontier	kg:read, kg:propose, classics:read	kg.term, kg.propose_change, bridge.phenotype, bridge.map, bridge.reverse	MappingProposals	40 · 6
pattern_miner	TaoChronos-PatternMiner	Pattern Miner. Tool-first discovery engines D1–D5 and source rediscovery; no model.	procedure	small	analysis:run, kg:read, classics:read	analysis.*, study.pairs, study.network	ObservationSet	60 · 0
philologist	TaoChronos-Philologist	Philologist (校勘). Readings, variants, collation status — never a forced single reading.	hybrid	medium	classics:read	philology.*, classics.get_passage, validation.verify_quote, study.variants, study.stemma, study.layers, study.dating, study.authorship, study.witnesses, study.punctuate, study.commentaries, study.glosses, study.clause, study.entities, study.translate, vision.layout, vision.compare	PhilologyReport	60 · 8
scholar	TaoChronos-Scholar	Source Scholar (治学). The textual history of the focus formulas, drugs and terms over the whole corpus.	procedure	small	classics:read	study.formula, study.herb, study.term, study.senses, study.concordance, study.taboo, study.metrology, study.commentaries, study.disputes, study.network, study.schools, study.names, study.places, study.translate	SourcesDossier	24 · 0
semanticist	TaoChronos-Semanticist	Historical Semanticist. Period-bound sense resolution; guards against homonymy.	hybrid	medium	classics:read, kg:read	kg.term, classics.get_passage, classics.search, study.term, study.senses, study.glosses, study.names, study.entities, study.translate, bridge.phenotype	TermResolutionReport	60 · 8
skeptic	TaoChronos-Skeptic	Skeptic (falsification). Tries to break every hypothesis before anyone believes it.	hybrid	frontier	classics:read, kg:read, analysis:run, codemode:exec	classics.search, classics.get_passage, classics.lineage, kg.claims, philology.assess_passage, validation.verify_quote, analysis.temporal_profile, study.reuse, study.dating, study.argument, study.disputes, audit.claim, graph.evidence	ReviewSet	400 · 12
statistician	TaoChronos-Statistician	Statistician. Significance, multiple-testing control and coverage for every observation; no model.	procedure	small	analysis:run	analysis.link_prediction, analysis.temporal_profile	AnalysisSet	20 · 0

Supplementary Table 5 | The tool mesh: families, the permission each needs and the tools (`tables/S5_tools.csv`)

Family	Permission needed	What the permission allows	Tools	Names
analytics	analysis:run	run the discovery and statistics engines	9	analysis.association_rules, analysis.communities, analysis.concept_drift, analysis.contradictions, analysis.formula_evolution, analysis.link_prediction, analysis.lost_knowledge, analysis.missing_sources, analysis.temporal_profile
bridge	classics:read	read the corpus through the research's scoped view	3	bridge.map, bridge.phenotype, bridge.reverse
knowledge	kg:propose	propose a change to canonical knowledge	1	kg.propose_change (writes)
knowledge	kg:read	read the knowledge graph	4	kg.claims, kg.extract_claims, kg.neighbors, kg.term
literature	literature:read	read modern literature and real-world data	4	literature.modern_evidence, modern.safety, modern.search, rwd.ask
philology	classics:read	read the corpus through the research's scoped view	6	philology.assess_passage, philology.collate, philology.inspect_page, philology.normalize, vision.compare, vision.layout
research	codemode:exec	run code in the sandbox	1	research.run_code
retrieval	classics:read	read the corpus through the research's scoped view	5	classics.canon, classics.get_passage, classics.lineage, classics.list_books, classics.search
study	classics:read	read the corpus through the research's scoped view	35	study.argument, study.authorship, study.cards, study.cases, study.chain, study.citations, study.clause, study.commentaries, study.concordance, study.dating, study.disputes, study.entities, study.formula, study.fragments, study.glosses, study.herb, study.layers, study.metrology, study.names, study.network, study.pairs, study.places, study.punctuate, study.reading, study.reuse, study.schools, study.senses, study.stemma, study.taboo, study.term, study.trajectories, study.translate, study.transmission, study.variants, study.witnesses
validation	classics:read	read the corpus through the research's scoped view	4	audit.claim, graph.evidence, validation.provenance_chain, validation.verify_quote
validation	validation:run	run the gates (validators)	1	validation.gates

Supplementary Table 6 | The hooks the harness runs and what each refuses (`tables/S6_hooks.csv`)

Hook point	Guard	Runs on	Refuses
BeforeToolCall	scope_guard	every tool call	arguments that name material outside the research's scope (withheld books, the future of a blind cutoff)
AfterToolCall	scope_firewall	every tool result	a result that carries withheld material back to the caller
BeforeClaimCommit	provenance_validator	every claim	a quote that is not verbatim, empty or outside its passage; inconsistent argument spans
BeforeClaimCommit	semantic_support	every claim	a claim its quote contradicts (a true quote with a false relation); frames the rest with speaker, stance, conditions and negations
BeforeClaimCommit	anachronism_guard	every claim	a classical claim that carries a modern concept
AfterHypothesis	anachronism_guard	every hypothesis and revision	a statement that equates a historical concept with a modern category, or breaks a forbidden assumption of the contract
BeforeGraphWrite	anachronism_guard	every mapping into the knowledge graph	a historical-to-modern equivalence asserted by anyone but a human expert
BeforeArtifactPublish	publish_guard	every report that is not a draft	a report without evidence, one that cites missing or retracted objects, or one that presents a

Hook point	Guard	Runs on	Refuses
			hypothesis not judged on its current version

Supplementary Table 7 | Event authority: the principals allowed to emit each event type (● allowed), with the reason where the table records one (`tables/S7_event_authority.csv`)

Event	Agent	Validator	Human	Kernel	Why
ResearchCreated				●	only the harness opens a research contract
ProfileLoaded				●	
CorpusScoped	●	●		●	
RoundStarted				●	
RoundCompleted				●	
StopConditionMet				●	
ResearchCompleted				●	only the harness completes a research
ResearchStopped			●	●	
ContractEvaluated		●		●	the research contract is evaluated by validators
TaskPlanned	●	●		●	
TaskStarted	●	●		●	
TaskCompleted	●	●		●	
TaskFailed	●	●		●	
TaskSkipped	●	●		●	
AgentSpawned				●	
AgentReleased				●	
ModelRouted				●	
LLMCalled	●			●	
ToolCalled	●	●	●	●	
ToolResult	●	●	●	●	
ToolDenied	●	●	●	●	
HookBlocked	●	●	●	●	
BudgetExceeded	●	●	●	●	
BudgetReserved				●	
BudgetSettled				●	
RunManifestRecorded				●	
PassageAssessed	●	●		●	
TermResolved	●	●		●	
EvidenceRetrieved	●	●		●	
EvidenceRetracted		●	●	●	
ClaimExtracted	●	●		●	
ClaimRejected	●	●		●	
LineageDetected	●	●		●	
ContradictionFound	●	●		●	
ObservationRecorded	●	●		●	
AnalysisRecorded	●	●		●	
WarrantRecorded	●	●		●	
HypothesisGenerated	●	●		●	
CounterEvidenceFound	●	●		●	
HypothesisReviewed	●	●		●	
HypothesisRevised	●	●		●	
HypothesisRejected		●	●	●	
HypothesisScored		●		●	scores are computed by validators
HypothesisRanked		●		●	rankings are computed by validators
HypothesisConfirmed		●		●	confirmation is a validator's re-test on withheld material
GateEvaluated		●		●	gates are graded by validators, never by the work they grade
ObjectionResolutionProposed	●	●		●	
ObjectionClosed		●	●		an objection is closed by re-running its check (validator)

Event	Agent	Validator	Human	Kernel	Why
					or by an expert, never by a model
ChangeProposed	•	•	•	•	
ChangeApproved		•	•		agents may only propose changes; a validator or human must decide
ChangeRejected		•	•		agents may only propose changes; a validator or human must decide
ExpertApproved			•		only a human expert may approve or reject a hypothesis (Gate G7)
ExpertRejected			•		only a human expert may approve or reject a hypothesis (Gate G7)
ExpertSteered			•		only a human expert steers a research
DecisionRecorded	•	•	•	•	
MemoryWritten		•	•	•	
BranchForked				•	
BranchMerged				•	
CheckpointCreated				•	
ArtifactPublished		•		•	publication is a validator's act after the contract is evaluated
TransactionCommitted				•	

Supplementary Table 8 | In conversation: the four core agents, the skills S01–S15 they run and the specialists they delegate to (`tables/S8_core_agents_skills.csv`)

Core agent	Skills	Specialists it delegates to	Permissions	Rules
古籍学者 Classical TCM Scholar	S01 来源与版本 Source & Provenance; S02 影像与版面 Document Vision; S03 句读·训诂 Philology; S04 条文结构 Structure & Clause; S05 校勘 Collation; S06 互文与传承 Intertext; S07 知识抽取 Knowledge Extraction; S08 名称规范 TCM Normalization; S09 源流演变 Historical Evolution	curator, philologist, extractor, semanticist, lineage, scholar, evolution	classics:read, kg:read	the original is never changed; every reading is a layer with its source / absence is only meaningful where the period's coverage is adequate
证据与批评 Evidence & Critic Agent	S15 证据审计 Evidence Audit	evidence, skeptic, statistician	classics:read, kg:read, literature:read, validation:run	a verdict is never raised by a model / SUPPORTED means the checks this system can run pass, not that the claim is true
研究编排者 Research Orchestrator	plan plan; synthesize synthesize	director, meta_reviewer	classics:read, kg:read, analysis:run	rules plan first; a model may add or drop optional skills, never S01 or S15, and never a skill the registry lacks / the answer says what the evidence says and no more; the critic's verdict is shown beside each statement
古今转译 Translational Medicine Agent	S10 古代表型化 Ancient Phenotyping; S11 本体桥接 Ontology Bridge; S12 古今映射 Clinical Mapping; S13 现代证据 Modern Evidence; S14 真实世界验证 RWD Validation	ontologist, modern_evidence, hypothesis	classics:read, kg:read, literature:read	phenotypic similarity, never retrospective diagnosis; no ancient name is equated with a modern disease / animal, cell and computational work is not efficacy in people; no clinical or dosage advice

Supplementary Note 3 | The end-to-end research case

The question '肾气丸的组成与主治如何随时代演变？ was put to the harness on the open canon corpus with `taochronos research "肾气丸的组成与主治如何随时代演变？" --profile web-canon --focus 肾气丸,六味地黄丸 --forbid 消渴=糖尿病`, offline: the rules ran alone and no model was called. The run scoped 206 books and 1,399 passages, planned 30 tasks in three rounds, called 156 tools (none refused), extracted 777 claims, recorded 87 evidence and 22 counter-evidence records and 24 hypotheses, and stopped after round 2, the last the contract allowed, when new evidence had fallen to 4.6%, below the 5% of its stop rule. It is deterministic: a second run writes the same report. The contract's verdict: one hypothesis (H1) meets every criterion a machine can check and waits for an expert, twelve are inconclusive and one (R1) is rejected. Supplementary Data 1 is the whole Discovery Report the run wrote, in Chinese, as a user of the harness receives it; Supplementary Data 2 is its event log, from which the state replays;

Supplementary Tables 9 and 10 list every hypothesis and objection. Reading each surviving hypothesis against its evidence found five failure modes (Supplementary Fig. 1), the largest being that the independence check merged 35 of the 38 books that supplied evidence into one source family.

Supplementary Table 9 | Every hypothesis of the end-to-end run (Fig. 3), with its gates, unmet criteria, objections and testable prediction (`tables/S9_e2e_hypotheses.csv`)

Label	Id	Kind	Generation	Parent	Status	Verdict	Gates G0–G8	Criteria unmet	Objectio (closed)
H1	hyp_d1436acd2959	association_rule	0		survived	needs_expert	G0:pass G1:warn G2:warn G3:pass G4:pass G5:pass G6:pass G7:pending G8:not_applicable	expert_review	0 (0)
H2	hyp_12a32285d3a3	lost_knowledge	1	hyp_71c99bcff38f	survived	inconclusive	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	6 (6)
H3	hyp_e2ebd991ae10	lost_source	0		survived	inconclusive	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	2 (0)
H4	hyp_24368056df5c	concept_drift	1	hyp_53c77f1a68e3	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	8 (8)
H5	hyp_3849a6721ca0	concept_drift	1	hyp_1ebef0b387ae	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	8 (8)

Label	Id	Kind	Generation	Parent	Status	Verdict	Gates G0–G8	Criteria unmet	Objectic (closed)
H6	hyp_de35c2a355f0	lost_knowledge	1	hyp_44172aa26142	survived	inconclusive	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	10 (10)
H7	hyp_0af4af4a83db	association_rule	1	hyp_dc14fa336232	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	semantic_support, source_dependence, current_version_gates, expert_review	2 (2)
H8	hyp_5730b22d8213	lost_source	0		survived	inconclusive	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:pass G7:pending G8:not_applicable	min_supporting_evidence, source_dependence, current_version_gates, expert_review	2 (0)
H9	hyp_ed4b5625f057	concept_drift	1	hyp_32bbf0aeb230	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	semantic_support, source_dependence, current_version_gates, expert_review	10 (10)

Label	Id	Kind	Generation	Parent	Status	Verdict	Gates G0–G8	Criteria unmet	Objectic (closed)
H10	hyp_502e05814d54	contradiction	1	hyp_539ab225761b	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	semantic_support, source_dependence, current_version_gates, expert_review	8 (4)
H11	hyp_18f4717f2182	contradiction	1	hyp_01f85940a40d	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	semantic_support, source_dependence, current_version_gates, expert_review	8 (4)
H12	hyp_80883924c519	hidden_association	1	hyp_412994ed6551	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	source_dependence, current_version_gates, expert_review	6 (6)
H13	hyp_7c62e0a0d161	hidden_association	1	hyp_8bec39add14d	survived	inconclusive	G0:pass G1:warn G2:warn G3:warn G4:pass G5:warn G6:pass G7:pending G8:not_applicable	semantic_support, source_dependence, current_version_gates, expert_review	6 (6)
R1	hyp_ccbe3e9607db	lost_knowledge	0		rejected	rejected	G0:pass G1:warn G2:warn G3:pass G4:pass G5:warn G6:fail G7:pending G8:not_applicable		5 (0)

Label	Id	Kind	Generation	Parent	Status	Verdict	Gates G0–G8	Criteria unmet	Objectic (closed)
(superseded)	hyp_01f85940a40d	contradiction	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		4 (0)
(superseded)	hyp_1ebef0b387ae	concept_drift	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		4 (0)
(superseded)	hyp_32bbf0aeb230	concept_drift	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		5 (0)
(superseded)	hyp_412994ed6551	hidden_association	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		3 (0)
(superseded)	hyp_44172aa26142	lost_knowledge	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		5 (0)
(superseded)	hyp_539ab225761b	contradiction	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		4 (0)
(superseded)	hyp_53c77f1a68e3	concept_drift	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		4 (0)

Label	Id	Kind	Generation	Parent	Status	Verdict	Gates G0–G8	Criteria unmet	Objectic (closed)
(superseded)	hyp_71c99bcff38f	lost_knowledge	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		3 (0)
(superseded)	hyp_8bec39add14d	hidden_association	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		3 (0)
(superseded)	hyp_dc14fa336232	association_rule	0		superseded		G0:· G1:· G2:· G3:· G4:· G5:· G6:· G7:pending G8:·		1 (0)

Supplementary Table 10 | The Skeptic's 117 objections in the end-to-end run, by check (each objection is in the CSV file) (`tables/S10_e2e_objections.csv`)

Check	Raised	Closed after narrowing	Open: major	Open: minor	Open: critical
single_source	35	16	9	10	0
transcription_dependence	29	18	11	0	0
statistical	15	10	5	0	0
homonym	9	6	3	0	0
sample_size	9	6	3	0	0
school	6	0	0	6	0
counterexample	6	4	2	0	0
earliest_source	4	2	2	0	0
partial_persistence	3	2	1	0	0
restated_after	1	0	0	0	1

Supplementary Note 4 | Prediction lists

Two lists record what the harness predicted before it could see the outcome, on the open canon corpus. Supplementary Table 11 lists the 82 predictions of the whole pipeline run on the past view at 1368 for two development topics (消渴, 中风) and four held-out topics (胸痹, 霍乱, 虚劳, 痰饮), each with its outcome in the later books (33 could be judged by the benchmark; formula evolutions, contradictions and lost books are not). No passage outside the past view was read (leakage 0). Supplementary Table 12 lists, for each of the 30 topic–cutoff pairs in which a link came true (56 links among 6,178 candidates), the average precision of three rankers and the true links Adamic–Adar placed in its top 20. The full-corpus runs of Fig. 5 (189 pairs, 1,314 links) need the store that is not distributed; their aggregates are in docs/evals.md.

Supplementary Table 11 | The predictions of the pipeline run on the past view at 1368 on the open corpus, and their outcomes (`tables/S11_predictions_open_corpus.csv`)

Topic	Split	Hypothesis	Kind	Review status	Prediction	Subject	Object	Held (strict)	Held (loose)	Later books	Note
disease: 消渴	dev	hyp_ec73bbdaf1a2	hidden_association	rejected	link_appears	symptom:咳	formula:五苓散	no	no	1	already written before the cutoff
disease: 消渴	dev	hyp_af6d5b43f7af	lost_source	survived	link_appears	work:wupu_bencao		not judged	not judged		not judged by this benchmark
disease: 消渴	dev	hyp_651fa1673f25	concept_drift	survived	sense_shift	disease:消渴	消渴#病·多饮多尿	yes	yes		

Topic	Split	Hypothesis	Kind	Review status	Prediction	Subject	Object	Held (strict)	Held (loose)	Later books	Note
disease:消渴	dev	hyp_7a1ec44e1bfd	association_rule	survived	cooccurrence_after	symptom:头痛,symptom:无大热	formula:越婢汤	not judged	not judged		
disease:消渴	dev	hyp_4ad17f25a523	lost_knowledge	survived	absent_after	formula:五苓散	disease:消渴	no	no	4	
disease:消渴	dev	hyp_2f86959af8d4	lost_knowledge	survived	absent_after	etiology:寒	disease:消渴	yes	no	0	
disease:消渴	dev	hyp_dc5e204b197e	lost_source	survived	link_appears	work:jingxin_lu		not judged	not judged		not judged by this benchmark
disease:消渴	dev	hyp_ae44900af968	contradiction	survived	resolution	disease:伤寒	contradiction	not judged	not judged		not judged by this benchmark
disease:消渴	dev	hyp_9f1c3f0ed371	formula_evolution	survived	herb_retained	herb:猪苓,herb:白术,herb:茯苓		not judged	not judged		not judged by this benchmark
disease:消渴	dev	hyp_823d383ae14e	contradiction	survived	resolution	disease:伤寒	contradiction	not judged	not judged		not judged by this benchmark
disease:消渴	dev	hyp_b4c92c32c385	contradiction	survived	resolution	disease:伤寒	contradiction	not judged	not judged		not judged by this benchmark
disease:消渴	dev	hyp_72149189a2f4	association_rule	survived	cooccurrence_after	symptom:不渴,symptom:头痛	formula:越婢汤	no	no		
disease:消渴	dev	hyp_514347a5ab71	hidden_association	survived	link_appears	symptom:小便不利	formula:麻黄汤	no	no	0	
disease:消渴	dev	hyp_08b2269de51a	lost_knowledge	survived	absent_after	etiology:服石	disease:消渴	yes	no	0	
disease:中风	dev	hyp_b556c451a515	lost_knowledge	rejected	absent_after	etiology:风	disease:中风	no	no	8	
disease:中风	dev	hyp_6391fe879596	lost_source	survived	link_appears	work:yingru_fang		not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_3495600970d8	contradiction	survived	resolution	pathogenesis:里寒	contradiction	not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_194ef9aa2f95	contradiction	survived	resolution	disease:中风	conditional	not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_373d31ba6e0d	concept_drift	survived	sense_shift	disease:中风		not judged	not judged		
disease:中风	dev	hyp_aa95d8eaed27	association_rule	survived	cooccurrence_after	pulse:微,symptom:头痛	formula:鸡屎白散	no	no		
disease:中风	dev	hyp_89303484a9d1	formula_evolution	survived	herb_retained	herb:大枣,herb:甘草,herb:生姜,herb:芍药		not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_51f54da65656	lost_source	survived	link_appears	work:玉函		not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_d83e15e95ac8	lost_knowledge	survived	absent_after	formula:五苓散	disease:中风	no	no	6	
disease:中风	dev	hyp_e0c76e58ffd1	formula_evolution	survived	herb_retained	herb:杏仁,herb:甘草,herb:石膏,herb:麻黄		not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_5ed6252ce76e	formula_evolution	survived	herb_retained	herb:甘草,herb:葛根,herb:黄芩		not judged	not judged		not judged by this benchmark
disease:中风	dev	hyp_1c0f22705407	association_rule	survived	cooccurrence_after	symptom:下利,symptom:半身不遂	formula:姜附汤	not judged	not judged		
disease:中风	dev	hyp_3bffbbea414e	hidden_association	survived	link_appears	symptom:喘	formula:小柴胡汤	no	no	0	
disease:中风	dev	hyp_4dbf0e894295	hidden_association	survived	link_appears	symptom:短气	formula:小柴胡汤	yes	yes	2	
disease:胸痹	heldout	hyp_4af9e0e3e79c	lost_source	survived	link_appears	work:gujin_luyan		not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_5ebe75623097	formula_evolution	survived	herb_retained	herb:栝楼实,herb:薤白		not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_a43b2b13831c	formula_evolution	survived	herb_retained	herb:甘草,herb:白术,herb:茯苓		not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_7b3550c17534	association_rule	survived	cooccurrence_after	pulse:微	formula:附子汤	no	no		
disease:胸痹	heldout	hyp_0910282fd418	contradiction	survived	resolution	disease:胸痹	contradiction	not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_fa95353f8b7c	contradiction	survived	resolution	disease:胸痹	contradiction	not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_d05ebd4bb9c5	contradiction	survived	resolution	disease:胸痹	contradiction	not judged	not judged		not judged by this benchmark

Topic	Split	Hypothesis	Kind	Review status	Prediction	Subject	Object	Held (strict)	Held (loose)	Later books	Note
disease:胸痹	heldout	hyp_664857ebf4f8	contradiction	survived	resolution	disease:胸痹	contradiction	not judged	not judged		not judged by this benchmark
disease:胸痹	heldout	hyp_ae095c1a8d06	association_rule	survived	cooccurrence_after	pulse:微,symptom:拘急	formula:附子汤	no	no		
disease:胸痹	heldout	hyp_a5b35de6aeeb	hidden_association	survived	link_appears	symptom:咳	formula:薤白汤	no	no	0	
disease:胸痹	heldout	hyp_6d7c0046ef32	hidden_association	survived	link_appears	disease:胸痹	herb:栝楼实	no	no	0	
disease:胸痹	heldout	hyp_90aded69eb74	concept_drift	survived	sense_shift	disease:胸痹		not judged	not judged		
disease:霍乱	heldout	hyp_1101b63a75f5	hidden_association	rejected	link_appears	disease:少阴病	formula:五苓散	no	no	0	
disease:霍乱	heldout	hyp_a640f4ce86db	concept_drift	survived	sense_shift	disease:霍乱		not judged	not judged		
disease:霍乱	heldout	hyp_64658eb3b657	hidden_association	rejected	link_appears	symptom:心烦	formula:四逆汤	no	no	0	
disease:霍乱	heldout	hyp_fab01fabcef8	lost_source	survived	link_appears	work:廣志		not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_631d624ae984	association_rule	survived	cooccurrence_after	pulse:细,symptom:心烦	formula:橘皮汤	not judged	not judged		
disease:霍乱	heldout	hyp_01ac93cc8c2c	lost_source	survived	link_appears	work:jinxiao_fang		not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_24329942a5fb	contradiction	survived	resolution	disease:伤寒	contradiction	not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_e2d7d528bdc3	lost_knowledge	survived	absent_after	etiology:生冷	disease:霍乱	yes	no	0	
disease:霍乱	heldout	hyp_6a1c6ff4847d	lost_knowledge	survived	absent_after	etiology:湿	disease:霍乱	no	no	3	
disease:霍乱	heldout	hyp_fb3ec7b257f7	formula_evolution	survived	herb_retained	herb:桂心,herb:泽泻,herb:猪苓,herb:白术,herb:茯苓		not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_5ddb1771ada9	contradiction	survived	resolution	disease:伤寒	contradiction	not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_63d3fa4367b7	association_rule	survived	cooccurrence_after	pulse:弦,symptom:心烦	formula:橘皮汤	not judged	not judged		
disease:霍乱	heldout	hyp_6bbe7b001ba1	formula_evolution	survived	herb_retained	herb:桔梗,herb:白术,herb:葛根		not judged	not judged		not judged by this benchmark
disease:霍乱	heldout	hyp_cbd5c278d27e	formula_evolution	survived	herb_retained	herb:地黄,herb:桂心,herb:白芍药		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_6e2c442d68f4	lost_source	survived	link_appears	work:gujin_luyan		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_eb599602bc5b	hidden_association	rejected	link_appears	disease:虚劳	formula:阳旦汤	no	no	0	
disease:虚劳	heldout	hyp_9bf01dd41feb	concept_drift	survived	sense_shift	disease:虚劳		not judged	not judged		
disease:虚劳	heldout	hyp_71c8050c5b39	contradiction	survived	resolution	symptom:喘	contradiction	not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_7f5d2b067426	lost_source	survived	link_appears	work:虚勞脉證第六		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_ef9914bc5ca5	hidden_association	rejected	link_appears	disease:中风	formula:桂枝汤	yes	yes	2	already written before the cutoff
disease:虚劳	heldout	hyp_e83081eba4fd	formula_evolution	survived	herb_retained	herb:人参,herb:地黄,herb:茯苓		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_fe67ebeb0ff0	formula_evolution	survived	herb_retained	herb:姜		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_757226e1d312	contradiction	survived	resolution	disease:虚劳	contradiction	not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_23d926167618	association_rule	survived	cooccurrence_after	symptom:心烦,symptom:鼻衄	formula:小建中汤	not judged	not judged		
disease:虚劳	heldout	hyp_8146f24b4b41	lost_knowledge	survived	absent_after	pathogenesis:痰	disease:虚劳	no	no	2	
disease:虚劳	heldout	hyp_ce997dbb3574	formula_evolution	survived	herb_retained	herb:人参,herb:白术,herb:陈皮		not judged	not judged		not judged by this benchmark
disease:虚劳	heldout	hyp_35e9d86013f7	lost_knowledge	survived	absent_after	formula:桂枝汤	disease:虚劳	yes	yes	0	

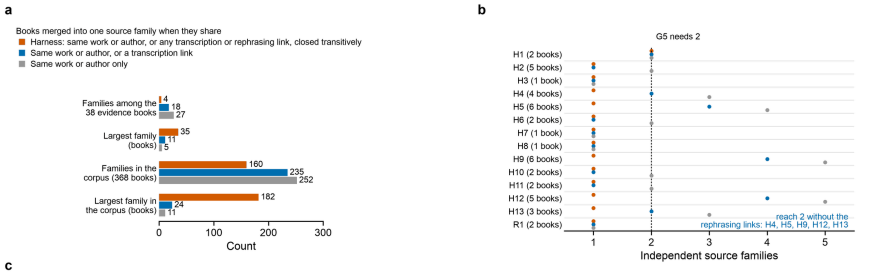
Topic	Split	Hypothesis	Kind	Review status	Prediction	Subject	Object	Held (strict)	Held (loose)	Later books	Note
disease:虚劳	heldout	hyp_daaa09c2954a	association_rule	survived	cooccurrence_after	pulse:微,symptom:拘急	formula:附子汤	no	no		
disease:痰饮	heldout	hyp_6e2c442d68f4	lost_source	survived	link_appears	work:gujin_luyan		not judged	not judged		not judged by this benchmark
disease:痰饮	heldout	hyp_5299b6d78bc6	hidden_association	rejected	link_appears	symptom:下利	formula:甘草汤	no	no	0	
disease:痰饮	heldout	hyp_5730b22d8213	lost_source	survived	link_appears	work:jiyan_fang		not judged	not judged		not judged by this benchmark
disease:痰饮	heldout	hyp_a5e57687fd43	hidden_association	rejected	link_appears	symptom:呕吐	formula:泽泻汤	no	no	0	
disease:痰饮	heldout	hyp_40c55f5cfa8a	formula_evolution	survived	herb_retained	herb:厚朴,herb:葛根,herb:陈皮		not judged	not judged		not judged by this benchmark
disease:痰饮	heldout	hyp_3196175ee3f3	association_rule	survived	cooccurrence_after	pulse:微,symptom:拘急	formula:附子汤	no	no		
disease:痰饮	heldout	hyp_50b7cabffc3b	concept_drift	survived	sense_shift	disease:痰饮		not judged	not judged		
disease:痰饮	heldout	hyp_0ccd266e27f9	association_rule	survived	cooccurrence_after	pulse:微	formula:附子汤	no	no		
disease:痰饮	heldout	hyp_db0d098c3ba7	lost_knowledge	survived	absent_after	etiology:寒	disease:痰饮	no	no	2	
disease:痰饮	heldout	hyp_c4211fadc6e5	lost_knowledge	survived	absent_after	formula:小半夏汤	disease:痰饮	yes	yes	0	
disease:痰饮	heldout	hyp_d4166477be17	concept_drift	survived	sense_shift	disease:伤寒	伤寒#广义	no	no		
disease:痰饮	heldout	hyp_992d7302feb3	formula_evolution	survived	herb_retained	herb:人参,herb:甘草,herb:白术		not judged	not judged		not judged by this benchmark
disease:痰饮	heldout	hyp_20f5f4132d38	concept_drift	survived	sense_shift	disease:中风	风#外风	no	no		
disease:痰饮	heldout	hyp_c2c854128b6c	contradiction	survived	resolution	disease:痰饮	contradiction	not judged	not judged		not judged by this benchmark

Supplementary Table 12 | Topic–cutoff pairs of the open corpus in which a link came true: average precision of three rankers and the true links Adamic–Adar ranked in its top 20 (`tables/S12_anticipated_links_open_corpus.csv`)

Cutoff	Topic	Past passages	Future passages	Candidates	Came true	AP Adamic–Adar	AP popularity	AP chance	First true link (Adamic–Adar rank)	True links in Adamic–Adar's top 20
1127	disease:中风	757	968	173	1	0.0667	0.0074	0.0243	15	formula:麻黄汤 (rank 15, 2 later books)
1127	disease:伤寒	1962	2752	231	6	0.0952	0.106	0.0441	4	herb:麻黄 (rank 4, 2 later books)
1127	symptom:下利	1145	1320	171	2	0.0355	0.0811	0.0318	22	
1127	symptom:下血	403	454	109	1	0.0122	0.013	0.0355	82	
1127	symptom:发热	1661	2767	163	3	0.0389	0.0172	0.0582	39	
1127	symptom:呕吐	854	1276	245	2	0.127	0.0103	0.0395	5	formula:五苓散 (rank 5, 4 later books)
1127	symptom:咳	1727	1783	227	1	0.1667	0.1111	0.0173	6	herb:杏仁 (rank 6, 2 later books)
1127	symptom:喘	1461	1639	157	3	0.0592	0.123	0.0436	19	herb:人参 (rank 19, 2 later books)
1127	symptom:头痛	974	1318	144	2	0.135	0.0201	0.0343	4	formula:四逆汤 (rank 4, 3 later books)
1127	symptom:寒热	1715	1800	250	1	0.0069	0.0048	0.0204	145	
1127	symptom:小便不利	569	514	83	1	0.0417	0.0139	0.0898	24	
1127	symptom:心烦	665	547	139	1	0.0213	0.0108	0.0246	47	
1127	symptom:无汗	247	534	18	1	0.0588	0.0667	0.1659	17	formula:茵陈汤 (rank

Cutoff	Topic	Past passages	Future passages	Candidates	Came true	AP Adamic–Adar	AP popularity	AP chance	First true link (Adamic–Adar rank)	True links in Adamic–Adar's top 20
										17, 2 later books)
1127	symptom:汗出	2217	1914	207	1	0.0185	0.0062	0.0249	54	
1127	symptom:浊	391	976	91	1	0.025	0.0233	0.0732	40	
1127	symptom:渴	1654	2732	224	4	0.0234	0.0166	0.0273	65	
1127	symptom:腹满	1004	711	166	3	0.1417	0.0151	0.0699	3	formula:桂枝汤 (rank 3, 2 later books)
1127	symptom:谵语	316	402	34	1	0.0435	0.0588	0.1126	23	
1368	disease:中风	1089	616	266	1	0.25	0.0147	0.0104	4	formula:麻黄汤 (rank 4, 2 later books)
1368	disease:伤寒	2811	1856	308	2	0.0361	0.0164	0.039	27	
1368	disease:疮	3404	2082	425	1	0.0132	0.0033	0.0111	76	
1368	symptom:发热	2369	2083	262	3	0.0202	0.0193	0.053	53	
1368	symptom:喘	2117	1070	309	2	0.2588	0.2539	0.0208	2	herb:人参 (rank 2, 2 later books)
1368	symptom:头痛	1468	843	213	1	0.0075	0.0069	0.0107	134	
1368	symptom:寒热	2334	1222	314	1	0.0123	0.0035	0.0122	81	
1368	symptom:汗出	2896	1263	281	1	0.0112	0.0044	0.0175	89	
1368	symptom:浊	626	727	142	1	0.0204	0.0139	0.0274	49	
1368	symptom:淋	898	619	231	1	0.0078	0.005	0.0121	128	
1368	symptom:渴	2567	1814	329	5	0.0159	0.0116	0.0396	122	
1368	symptom:腹满	1325	408	266	2	0.0387	0.0074	0.021	15	formula:桂枝汤 (rank 15, 2 later books)

Supplementary Figure



c

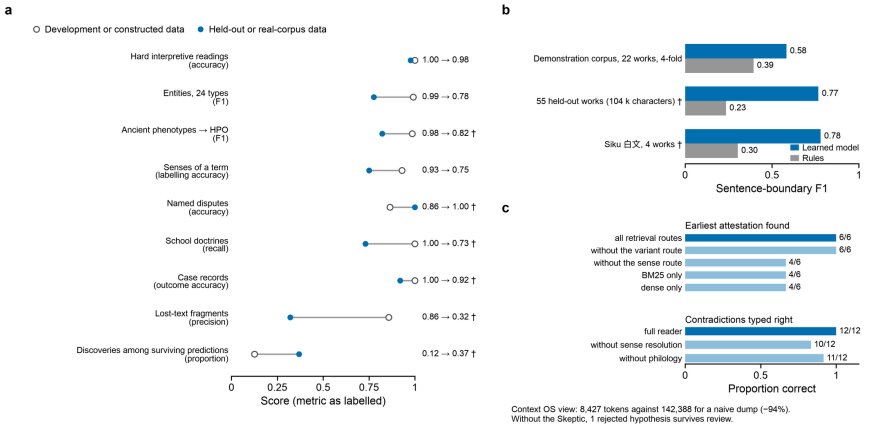
Failure	Where	What it did	What would catch it
Source families over-merged	12 of 13 candidates (Fig. 3d)	one rephrased passage merges two whole books, and the closure chains 182 of 368 books into one family; G5 then falls five hypotheses that rest on two to four families without the rephrasing links	judge independence per claim, from the passages that support it, not per book
Co-occurrence read as association	R1; H2	R1: 理中 and fever share a passage but not a prescription, in the Song text and in the later book alike; H2 reads 烦热 (vexing heat) as 心烦	require the relation to be stated (indication, not co-mention) before a lost or restated link counts
Commentary dated with its base text	H10, H11	the passages of 校注妇人良方 are Xue Ji's case notes (1547), dated with Chen Ziming's text (1237); the '301-year' contradiction is one physician against himself (Xue Ji also wrote 内科摘要)	mark editors' notes and case records as a later layer when the transcription does not
Duplicate statements	H10 = H11	two passage pairs of the same two books give one statement twice	merge hypotheses whose statements and terms coincide
Coverage of the open corpus	sources dossier	the open corpus keeps 8 of 24 chapters of 小儿药证直诀 (1119), none with 地黄丸, so the earliest composition of 六味地黄丸 found is of 1638	state the sampled share beside every earliest attestation; rerun on the full store

Also reported, not repaired: the two unsafe readings of the held-out interpretive cases (Fig. 4c) and the open-corpus benchmark, too small to separate the ranking from chance (Extended Data Fig. 2).

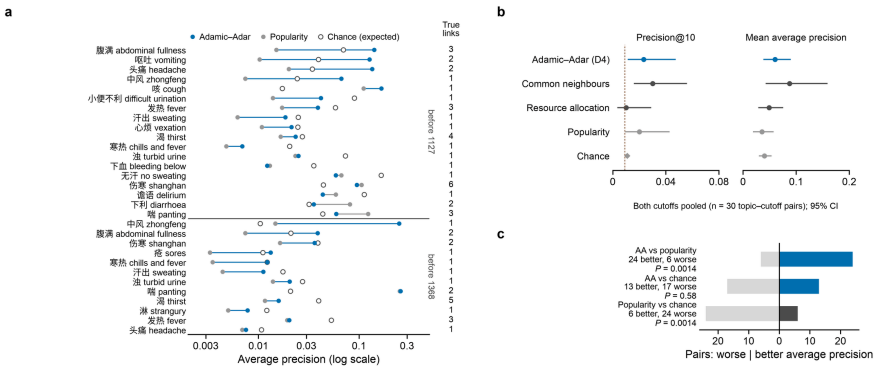
Supplementary Fig. 1 | Where the end-to-end run of Fig. 3 went wrong, and why. a, Books merged into one source family under three rules: the harness's (the same work or author, or any transcription or rephrasing link between their passages, closed transitively), transcription links only, and the same work or author only. Under the harness's rule 35 of the 38 books that supplied evidence, and 182 of the 368 books of the open corpus, fall into one family. **b**, The independent families each final hypothesis rests on under the three rules; dashed line, the two that G5 requires. Five hypotheses that G5 failed (H4, H5, H9, H12, H13) rest on two to four families once rephrasing links are left out. **c**, The failure modes found by reading

every surviving hypothesis against its evidence, with what each did and the check that would catch it. None is repaired here: the run is reported as it happened.

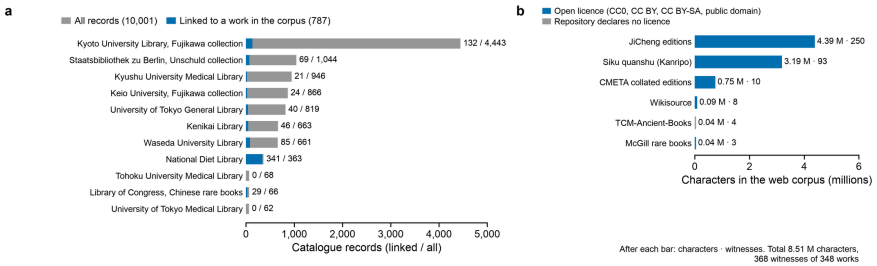
Supplementary Figures 2–4



Supplementary Fig. 2 | Where the rules transfer and where they do not. a, Development or constructed data (open circles) against held-out or real-corpus data (blue) for nine capabilities; numbers, the two scores; †, real-corpus value from the full corpus (`docs/evals.md`). Development sets were written with the rules they test. **b**, Sentence-boundary F1 of the learned punctuation model and of the rule baseline on the demonstration corpus (four folds), on 55 works held out of training, and on four unpunctuated *Siku* texts scored against their JiCheng punctuation (†, full corpus); every character was preserved in all settings. **c**, Ablations on the demonstration corpus: earliest-attestation accuracy with retrieval routes removed (6 terms), contradiction typing without sense resolution or philology (12 passage pairs), the context supplied to agents, and review without the Skeptic.



Supplementary Fig. 3 | The research-value benchmark on the open canon corpus. The benchmark of Fig. 5 re-run with identical rules on the open web corpus (348 works, most sampled by chapters). **a**, Average precision of Adamic–Adar (blue), popularity (grey) and the expectation under chance (open circles) in each of the 30 topic–cutoff pairs in which at least one link came true (18 at 1127, 12 at 1368; log scale); lines join Adamic–Adar to popularity, blue where Adamic–Adar is higher; right, links that came true in the pair. Most pairs have one or two true links (56 among 6,178 candidates), so a single hit moves a ranking. **b**, Precision@10 and mean average precision of the five rankers, both cutoffs pooled (n = 30 pairs); dots, means; lines, 95% confidence intervals; dashed line, base rate (0.91%). **c**, Paired comparisons of average precision, two-sided sign tests. Adamic–Adar still ranks above popularity (24 of 30 pairs, P = 0.0014) but cannot be told from chance (13 better, 17 worse, P = 0.58), and popularity falls below chance (6 better, 24 worse, P = 0.0014). The benchmark needs the full corpus; the replication is reported as a limit of the open subset. Every pair and every prediction: Supplementary Tables 11 and 12.



Supplementary Fig. 4 | The witnesses behind the texts. a, Catalogue records of printed and manuscript copies in eleven digitized collections (title, date, print or manuscript, shelfmark, IIIF manifest and licence; no image is downloaded) and those linked by title to a work of the corpus (787 records of 179 works); links made by title need checking by hand. **b**, Sources of the open web corpus in millions of characters and witnesses. Blue, openly licensed transcriptions: JiCheng, whose selection, arrangement, punctuation and notes on public-domain texts are released under CC0 1.0 and its own writing under CC BY 4.0 (the modern editors' part of the few texts it took from modern reconstructions of lost books excepted); CMETA, CC BY

4.0; Kanseki Repository and Wikisource, CC BY-SA 4.0; McGill, public domain. Grey, the four TCM-Ancient-Books texts that match no JiCheng text, whose repository declares no licence, carried on the decision of the releasing institute (IMPF-AI). Sources whose terms forbid redistribution are never included. IIIF, International Image Interoperability Framework.

Supplementary Table 13

Supplementary Table 13 | Sources of the corpus

Source	Licence	Works admitted	Characters (millions)	Transcription and notes
Kanseki Repository KR3e, Siku quanshu medical section (漢籍リポジトリ)	CC BY-SA 4.0	100	25.66*	Wenyuange Siku and Sibu congkan facsimiles; unpunctuated; catalogue with authors, dates, layers
JiCheng 笈成 edited texts, v1.4.8	CC0 1.0 (editorial work); base texts public domain§	816	93.64*	Volunteer transcriptions with modern punctuation; 41 of 857 works excluded
CMETA 數位中醫校書郎 collated editions	CC BY 4.0 (collation); base texts public domain	11	1.49	Collated page by page against images; passages carry page numbers and image links
Tōa Igaku Kyōkai 東亜医学協会 texts	© the society; local research only†	7	0.33	Suwen, Lingshu, Nanjing, Shanghan lun, Jingui yaolüe, Shennong bencao jing, Bian Que biography
Wikisource (Category 中醫, 醫書)	CC BY-SA 4.0	18	0.77	Works new after de-duplication of 627 candidates
McGill University Library rare books	Public domain	5	0.40	Qing prints and manuscripts on gynaecology
TCM-Ancient-Books; tcmoc	Not declared by the repositories‖	8	0.37	Mostly simplified copies of JiCheng texts, de-duplicated work by work
Hugging Face classical-tcm-canon	Proprietary-commercial; local research only†	1	0.31	One work (Zhang Zhicong's commentary on the Shanghan lun)
Kanseki Repository catalogue (KR-Catalog)	CC BY-SA 4.0	–	–	Roles and life dates of 132 persons responsible for the Siku works
Image catalogues of 11 collections‡	Recorded per collection	–	–	10,001 records (787 linked to 179 works): title, date, print or manuscript, shelfmark, IIIF manifest
Total		966	≈123	2.05 M passages; 761 distinct works; analysis periods 475 BCE–1949

*Counted in the corpus store (README); the catalogues of these two sources do not record characters. Other counts sum the catalogue records of the admitted works.

†Used for local research only and never redistributed; the open web corpus carries none of the texts whose terms forbid redistribution.

§JiCheng releases its selection, arrangement, punctuation and notes on public-domain texts under CC0 1.0 and its own writing under CC BY 4.0 (https://jicheng.tw/tcm/copyright.html, consulted 29 September 2026); the open web corpus carries 250 JiCheng witnesses under these terms, crediting JiCheng. The modern editors' part of the few texts JiCheng transcribed from modern reconstructions of lost books is not covered.

‖Mostly simplified-character copies of JiCheng texts; de-duplication removes every copy, and the 8 texts kept match no JiCheng text. The 4 in the open web corpus are carried on the decision of the releasing institute (IMPF-AI); the others are used for local research only.

‡Kyoto University (Fujikawa), Staatsbibliothek zu Berlin (Unschuld), Kyushu University, Keio University (Fujikawa), University of Tokyo (two libraries), Kenikai, Waseda University, National Diet Library, Tohoku University and Library of Congress. Only metadata are held; no image is downloaded.

Sources are ranked by trust for de-duplication (Kanripo, JiCheng, McGill, CMETA, AEAM, Wikisource, web collections); a candidate sharing enough normalized 12-character shingles with an admitted work is recorded as its copy, while McGill, CMETA and AEAM transcriptions are kept as independent witnesses. IIIF, International Image Interoperability Framework.

Supplementary Data

Supplementary Data 1 | The Discovery Report of the end-to-end run (`Supplementary_Data_1_e2e_discovery_report.md`): the report the harness wrote, in Chinese with English section titles, as the conversation client and the command line show it.

Supplementary Data 2 | The event log of the end-to-end run (`Supplementary_Data_2_e2e_event_log.jsonl.gz`): every event in order, one JSON object a line, each with its sequence number, type, actor, principal and payload; replaying it rebuilds the state from which Fig. 3 and the report were drawn.